

KRASNYY YAR, Russian Federation

WMO: 318450

Lat: 46.5404N

Lon: 135.3357E

Elev: 130

StdP: 99.77

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.7	-34.5	-39.4	0.1	-36.5	-37.2	0.1	-34.3	6.0	-12.8	4.9	-14.5	0.0	80	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	30.8	22.3	29.2	21.8	27.6	21.0	24.2	28.4	23.1	27.4	22.2	26.1	1.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.0	26.6	21.8	16.8	25.3	20.9	15.8	24.4	74.1	28.5	69.7	27.5	66.0	26.2	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.2	4.4	3.7	-39.7	33.9	2.7	1.2	-41.6	34.8	-43.2	35.5	-44.7	36.2	-46.6	37.1		
			-39.8	26.3	2.7	1.3	-41.7	27.2	-43.3	28.0	-44.8	28.7	-46.7	29.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.7	-21.5	-16.9	-7.5	3.6	11.9	17.6	21.4	20.1	13.3	4.2	-8.2	-19.2
	DBStd	15.81	5.17	5.30	6.31	4.42	3.77	3.41	2.50	2.95	3.80	4.44	6.79	5.59
	HDD10.0	4141	976	752	542	196	24	0	0	0	13	186	547	906
	HDD18.3	6290	1234	986	800	441	200	53	5	17	156	438	797	1164
	CDD10.0	1101	0	0	0	5	84	228	354	313	112	6	0	0
	CDD18.3	209	0	0	0	0	2	31	101	71	4	0	0	0
	CDH23.3	1953	0	0	0	13	117	451	822	494	54	2	0	0
	CDH26.7	553	0	0	0	2	24	141	265	113	7	0	0	0
Wind	WSAvg	1.2	0.9	1.1	1.5	1.5	1.5	1.1	1.0	0.9	1.0	1.3	1.4	1.0
Precipitation	PrecAvg	872	19	18	35	49	82	94	175	150	103	70	45	34
	PrecMax	1361	38	32	81	87	159	240	398	282	173	175	74	75
	PrecMin	613	3	3	3	14	21	42	43	39	39	18	12	7
	PrecStd	146	9	9	19	21	36	49	81	61	37	32	16	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.3	2.9	11.6	25.1	29.5	32.5	33.1	31.6	27.5	21.8	11.2	-0.9
		MCWB	-5.0	-0.3	4.9	12.8	17.9	22.0	23.7	23.8	20.6	13.1	5.9	-2.2
	2%	DB	-7.2	-0.5	7.7	20.4	26.4	30.1	31.2	29.4	24.7	17.9	6.7	-3.5
		MCWB	-8.4	-3.1	2.6	10.2	15.7	20.9	23.0	23.2	18.6	11.5	3.6	-4.5
	5%	DB	-9.9	-3.6	5.2	16.2	24.1	28.1	29.6	27.8	22.7	15.1	4.0	-6.5
		MCWB	-10.9	-5.5	1.0	8.0	14.5	19.9	22.6	22.2	17.3	9.5	2.0	-7.4
	10%	DB	-12.0	-6.1	3.0	13.2	21.4	25.9	27.9	26.2	20.7	12.5	1.4	-9.5
		MCWB	-12.8	-7.8	-0.1	6.6	13.3	19.0	21.9	21.5	16.4	8.3	-0.1	-10.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.8	0.1	5.4	13.3	18.2	23.6	25.6	25.6	21.7	14.1	6.7	-1.8
		MCDB	-3.5	2.5	10.4	24.3	27.5	30.3	29.9	29.3	25.2	19.4	10.3	-1.1
	2%	WB	-8.4	-3.1	3.1	10.7	16.8	21.8	24.4	24.2	19.7	12.3	4.1	-4.5
		MCDB	-7.4	-0.6	6.7	18.7	24.2	28.3	29.0	27.7	23.2	16.3	6.3	-3.5
	5%	WB	-10.8	-5.5	1.5	8.7	15.5	20.7	23.4	23.1	18.3	10.5	2.1	-7.4
		MCDB	-9.9	-3.6	4.7	15.1	22.1	26.6	28.0	26.7	21.5	14.1	3.9	-6.6
	10%	WB	-12.9	-7.8	0.0	7.0	14.1	19.6	22.5	22.1	16.9	8.7	0.1	-10.3
		MCDB	-12.1	-6.1	3.0	12.1	19.8	24.9	26.8	25.1	19.8	11.9	1.4	-9.5
Mean Daily Temperature Range	5% DB	MDBR	16.2	18.6	16.4	14.5	14.9	13.5	11.2	10.6	13.4	13.9	12.6	14.3
		MCDBR	17.1	20.4	17.8	21.9	21.3	18.6	15.2	13.4	16.7	19.3	15.0	13.7
		MCWBR	15.9	17.6	12.8	12.5	10.9	9.6	7.9	7.6	10.3	12.3	11.2	12.9
	5% WB	MCDBR	16.9	19.6	15.8	19.9	18.2	16.2	12.8	11.3	13.7	16.8	13.6	13.3
		MCWBR	15.7	17.0	11.6	11.8	9.8	8.9	7.4	7.0	8.9	11.3	10.5	12.4
Clear-Sky Solar Irradiance	taub		0.312	0.369	0.451	0.549	0.578	0.545	0.512	0.472	0.428	0.440	0.393	0.334
	taud		2.039	1.877	1.731	1.705	1.732	1.903	2.048	2.162	2.191	1.994	2.002	2.000
	Ebn at Noon		737	755	744	712	711	741	761	777	775	674	623	647
	Edh at Noon		99	145	196	221	223	189	162	139	123	129	104	93
All-Sky Solar Radiation	RadAvg		2.00	3.12	4.21	4.46	5.13	5.51	5.11	4.54	3.88	2.64	1.76	1.51
	RadStd		0.12	0.15	0.34	0.46	0.48	0.67	0.46	0.28	0.34	0.22	0.16	0.10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page